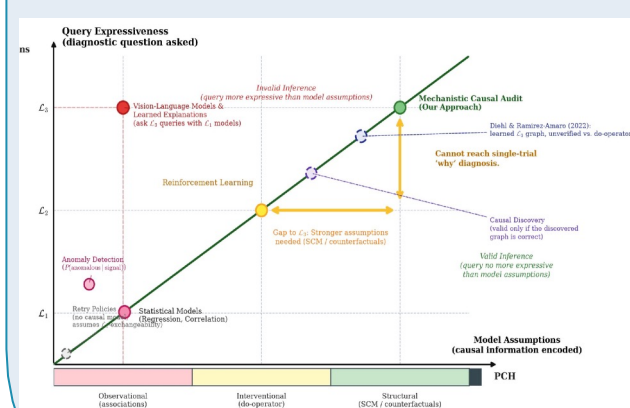


Counterfactual Diagnosis of Robotic Grasp Failures under Perceptual and Geometric Degradation

Problem: Robots don't know why they fail

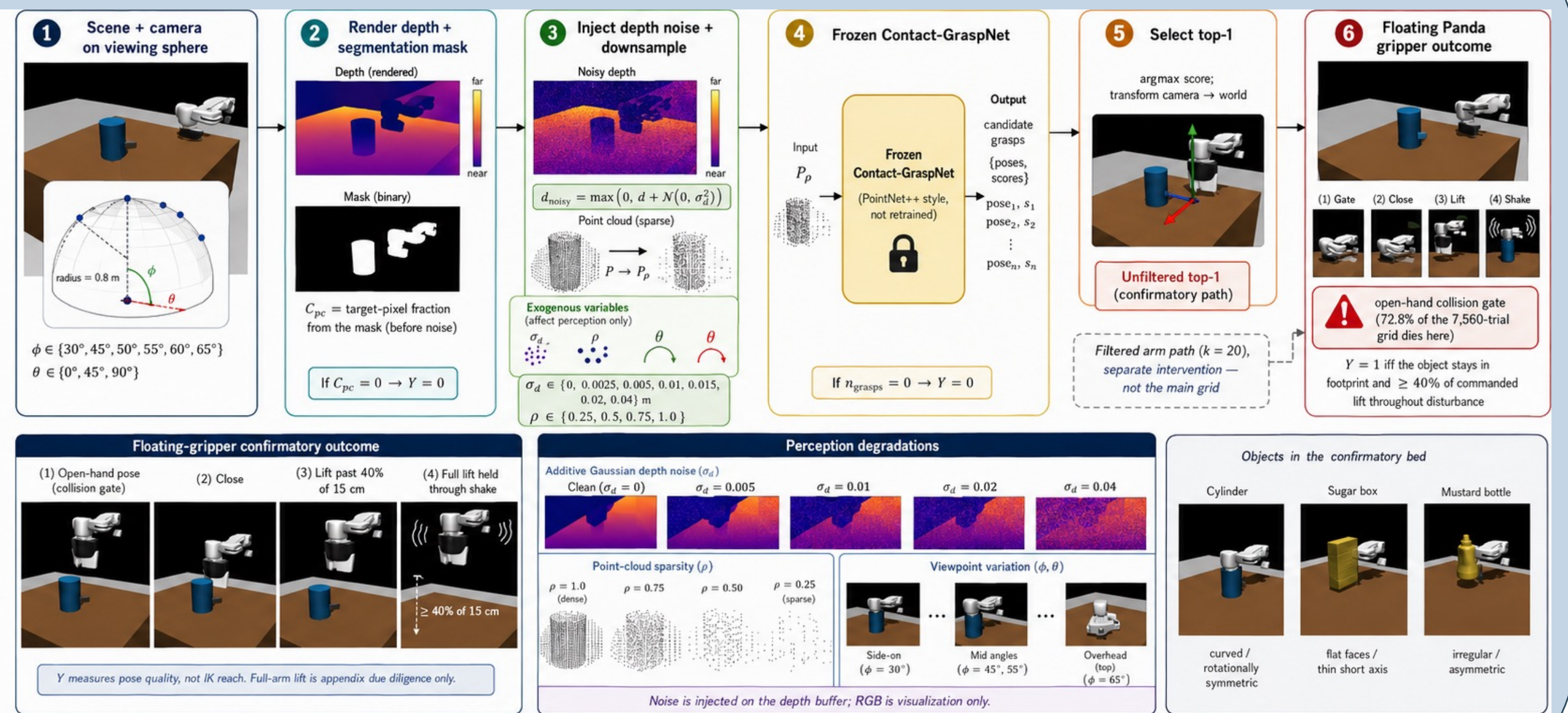
Grasp planners like **Contact-GraspNet** report **> 90% success** but only under lab conditions: clean sensing, favourable viewpoints and hidden collision filtering. In the real world, grasps fail from tangled combinations of **perception** (sensor noise), **geometry** (bad viewpoints) and **physics** (collisions).

When a grasp fails, the system only knows *that* it failed - not *why*.

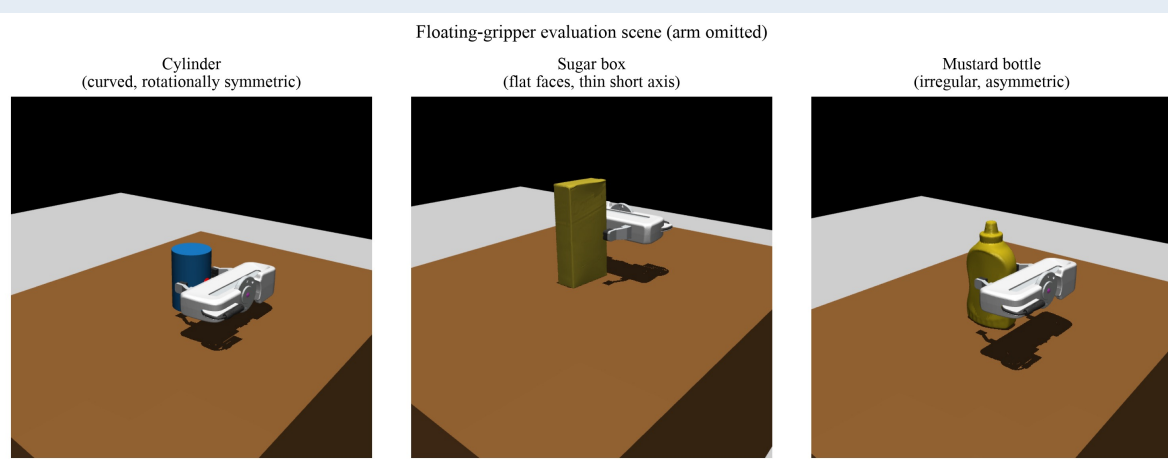


- Existing responses fall short:
 - Retry policies
 - Anomaly detection
 - LLM/VLM explanations
- The gap: no method combines (a) a structural model built from the robot's *known architecture* with (b) counterfactual predictions *checked against ground truth*.

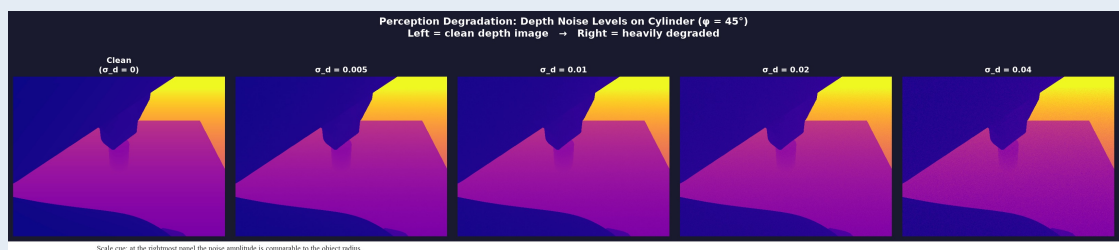
Method: A causal inference engine



Experiment Setup



Three geometries expose different contact and pose-selection failure modes.



Same simulated scene, progressively corrupted depth.



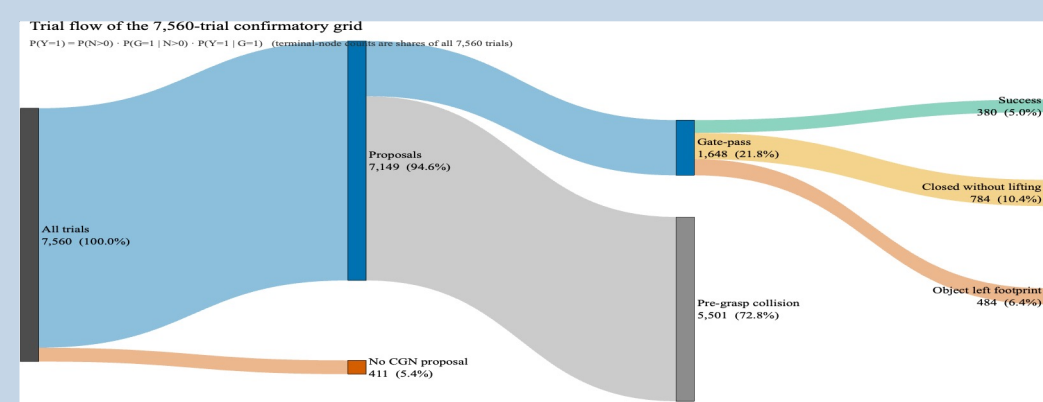
- 7,560 trials: 3 objects $\times$ 7 $\sigma_d \times$ 4 $\rho \times$ 6 $\phi \times$ 3 $\theta \times$ 5 seeds.
- Unfiltered top-1 pose; collision gate evaluated with the hand open.
- Physical outcome: close, lift 15 cm, and hold through a shake.
- Three-stage factorisation separates proposal, geometry, and execution.

Key Result 1: Geometry dominates failure, not perception

$$P(\text{success}) = P(\text{proposal}) \times P(\text{clears gate}) \times P(\text{holds lift}) = 5.0\%$$

94.6% 23.1% 23.1%

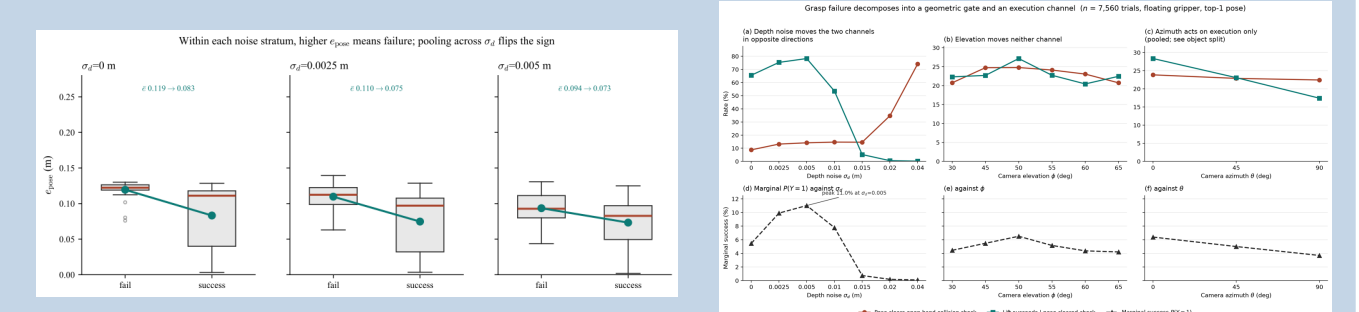
Protocol	Success	Pre-grasp collision
Top-1 only (no retry)	22/100 (22%)	78%
Top-20 + collision filter	89/100 (89%)	11%



Only 5.0% of 7,560 unfiltered top-1 grasps succeed and 72.8% never reach execution: the selected pose is already in collision.

Intervening on the gate (top-k collision filtering, k=20, clean viewpoint) lifts success to 89%. The same 100 trials under top-1 succeed only 22%.

Key Result 2: Correlation is not reliable



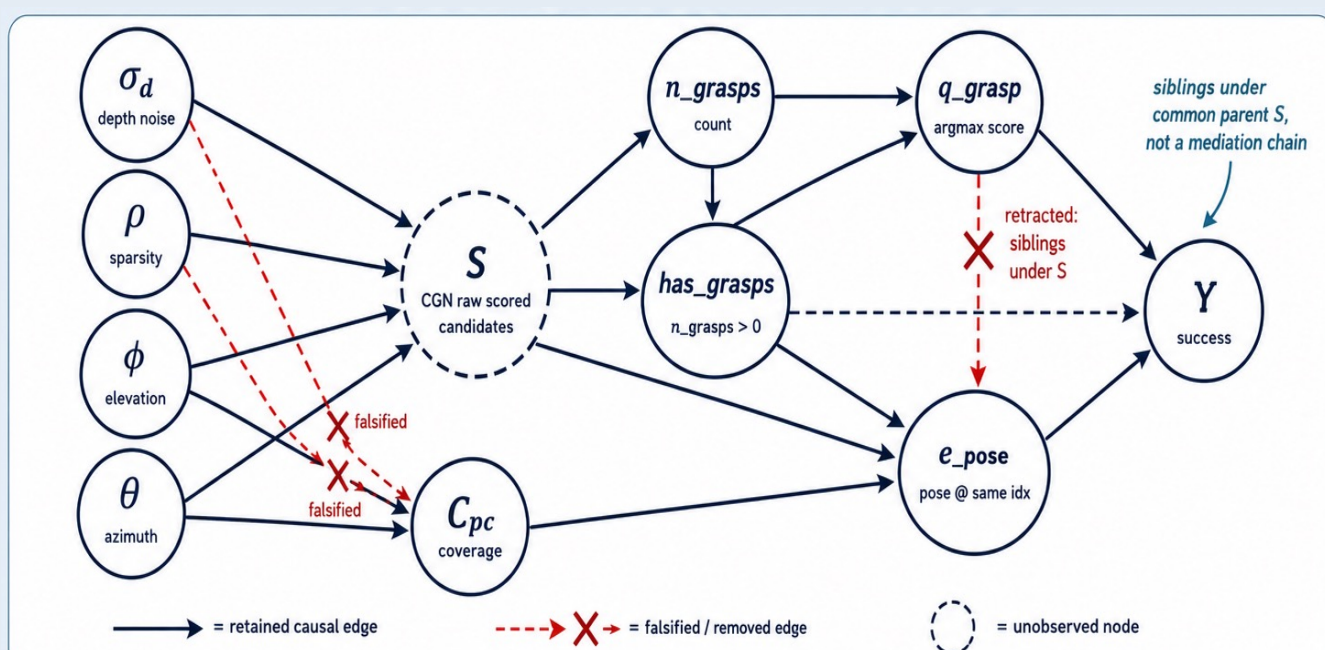
Depth noise moves the two channels in opposite directions: noise pushes proposals *off* the object, so more poses clear the collision gate (8.8% $\rightarrow$ 74.1%) even as post-gate success collapses (65.6% $\rightarrow$ 0.2%).

The marginal success rate therefore **peaks at mild noise** (11.0% at $\sigma_d = 0.005$).

Simpson's reversal on pose error: pooled across noise, *higher* pose error correlates with *success* (+0.25 on the cylinder). Within every noise stratum, the sign is correct.

Structural Causal Model

The causal graph is pre-registered from the pipeline's dataflow



1. Abduction

Recover the realised noise terms U .

2. Action

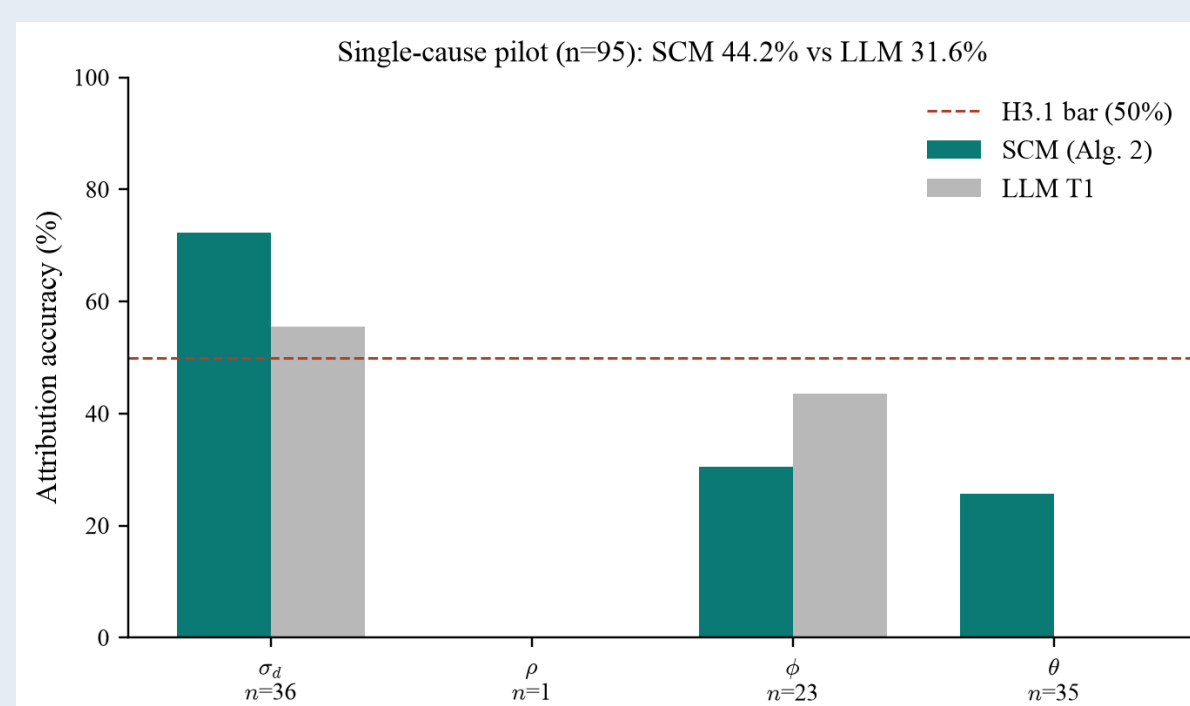
Replace one equation with $do(X=x')$.

3. Prediction

Propagate forward to counterfactual $Y_{x'}$.

A failure is attributed only when the single reset changes the predicted outcome; otherwise it remains joint or irreducible within the tested reset.

Diagnosis: SCM vs. zero-shot LLM



SCM: 42/95 = 44.2% | LLM: 30/95 = 31.6% | pre-registered 50% bar missed

On the 432-trials, we scored Pearl's counterfactual procedure against **re-simulated ground truth**

Method	σ_d (n=36)	ρ (n=1)	ϕ (n=23)	θ (n=35)	Mean (n=95)
Causal SCM (ours)	72.2	0.0	30.4	25.7	44.2
LLM baseline (T1)	55.6	0.0	43.5	0.0	31.6

The SCM beats the LLM but **misses our own pre-registered 50% bar**. The miss concentrates on viewpoint variables, matching a known Contact-GraspNet pose bias at $\phi=30^\circ$ that a linear surrogate can't capture. **57.2% of failures had no single-variable fix** (irreducibly multi-causal).

Takeaways

Grasp failure decomposes into a **geometric gate** and an **execution channel** - the marginal success rate confounds them. Pipeline structure known *by construction* beats both statistical discovery and associative LLM explanation and its failures are *locatable*, not hidden. Under full resets, **most failures are multi-causal** and "find the one bad variable" is often the wrong question.

Limits: simulation only; floating gripper $\neq$ full-arm pick rates; the collision gate is absent from the pre-registered graph; diagnostic scoring is on the pilot outcome.
Next: re-fit the SCM *with* the gate node on physical Y ; partial-observability abduction; VLM baseline with images.